# Building Management System

# Topics Covered:

a) Basics
b) Purpose
c) System Architecture
d) Controller I/O's distribution
e) Control Loops and Modes of Operation.
f) Interlocks
g) Power Failure Recovery
h) Alarms and Warnings
i) Trend Reporting

# BMS

- BMS stands for Building Management System
- BMS uses C-BUS, MOD-BUS, TCP/IP, BACNET,etc. communication protocols

- **BMS can be used for following systems**:

a) Heating ventilation and air conditioning
b) Walk in chamber
c) CCTV
d) Cold storage chamber
e) Door interlock system
f) FFU (Fan Filter Unit)
g) FAS ( Fire alarm system)

# Purpose

- To provide environmental control in Clean Room area that is in compliance with 21 CFR Part 11

# BMS Provides

- Monitoring
- Controlling
- Alarming
- Trending
- Reporting
- Archiving

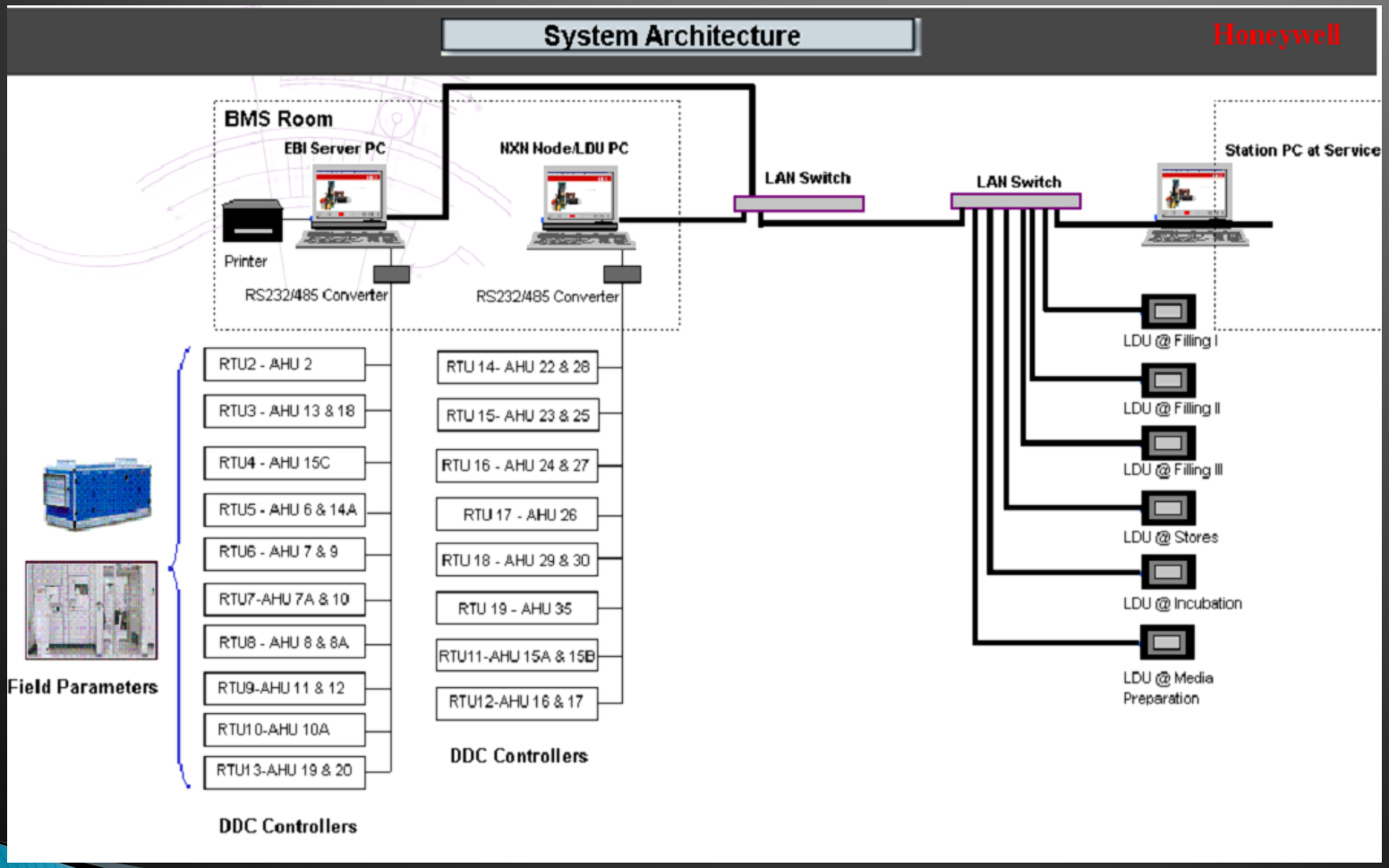
# Objectives And Benefits

- Standalone precision control of parameters
- Centralized monitoring and control
- Improve individual AHU efficiency

# Functions

- Audit trail for all user actions
- Data can be sampled and logged
- Modes of working
- Manual operation of equipment and device whenever required
- Use of high speed network

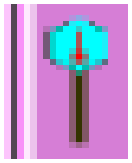
# System Architecture



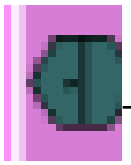
ROOM AREAS

# INPUT/OUTPUT DEVICES

DDC CONTROLLER IN BMS PANEL



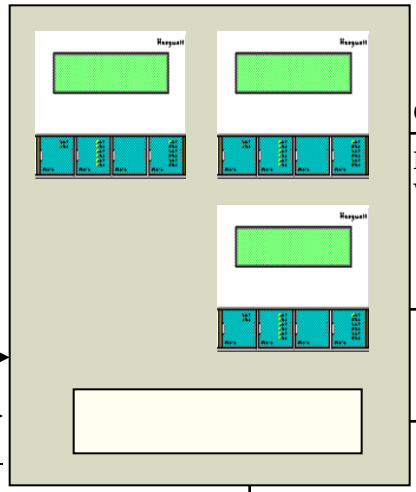
ROOM TEMPERATURE
SENSOR/TRANSMITTERS



ROOM PRESSURE TRANSMITTERS

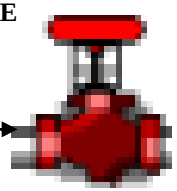


ROOM RH TRANSMITTERS

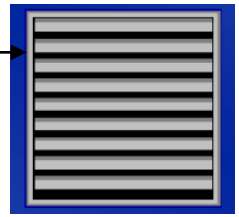




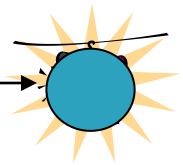
CONTROL VALVE
IN CHILLED
WATERRETURN LINE



CONTROL VALVE
IN HOT WATER
RETURN LINE

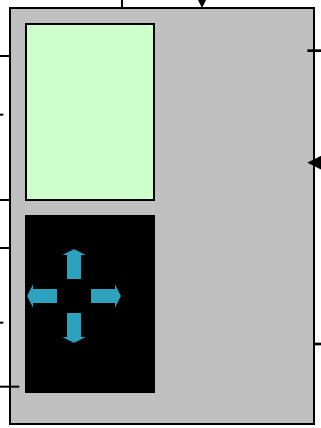


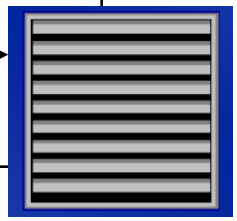
DAMPERS



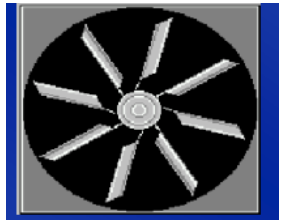
FLASHERS

AUTO MANUAL STATUS
VFD TRIP STATUS
CURRENT FEEDBACK
SPEED FEED BACK
AHU START STOP
VFD RESET





FIRE DAMPER



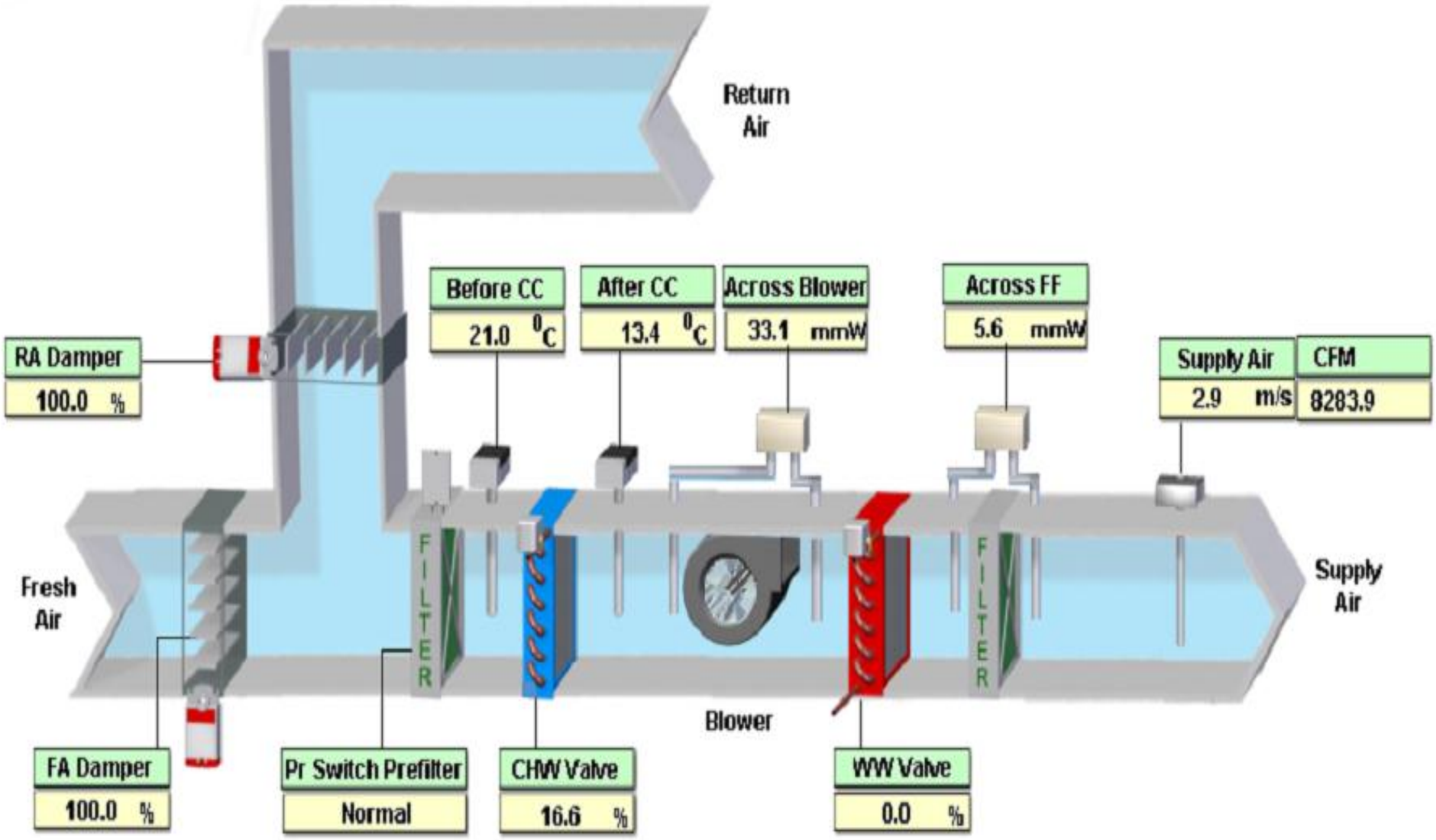
BLOWER SPPED CONTROL

VARIABLE FREQUENCY DRIVE (VFD)
IN AHU PANEL

# Interlocks

- AHU mode selection
- Time Program
- VFD trip status
- Fire damper status
- Door limit Switch status
- Defumigation Exhaust Damper actuator

# AHU DETAILS


Return Air
Before CC
21.0 °C
After CC
13.4 °C
Across Blower
33.1 mmW
Across FF
5.6 mmW
RA Damper
100.0 %
Supply Air
2.9 m/s
CFM
8283.9
Fresh Air
FILTER
FILTER
Supply Air
Blower
FA Damper
100.0 %
Pr Switch Prefilter
Normal
CHW Valve
16.6 %
WW Valve
0.0 %

# Modes of Operation

## Normal Mode:

| Sr. No. | Description | Output Value |
|---|---|---|
| 1 | Riser VCD | 4% to 90% Modulation |
| 2 | Defumigation Exhaust VCD | 0% OPEN |
| 3 | Fresh Air VCD | 0% To 100% Modulation |
| 4 | Main Return air VCD | 100% OPEN |
| 5 | Relief air VCD | 0% To 100 % modulation |
| 6 | Control valve actuator in chilled water line | 0% To 100 % modulation |
| 7 | Control valve actuator in warm water line | 0% To 100 % modulation |
| 8 | AHU blower Motor | Minimum to maximum speed |

# Defumigation Mode :

| Sr No. | Description | System component Status | | |
|---|---|---|---|---|
| | | Fumigation | Fumigation Hold | Defumigation |
| 1 | AHU | ON | OFF | ON |
| 2 | Fumigation Heater | ON | OFF | OFF |
| 3 | Fresh Air Supply VCD | 0% OPEN | 0% OPEN | 100% OPEN |
| 4 | Main Supply Air | 100% OPEN | 100% OPEN | 100% OPEN |
| 5 | Room RAVCD | 90% OPEN | 90% OPEN | 90% OPEN |
| 6 | Main return VCD | 100% OPEN | 100% OPEN | 0% OPEN |
| 7 | Defumigation Exhaust VCD | 0% OPEN | 0% OPEN | 100% OPEN |
| 8 | Relief Air VCD | 0% OPEN | 0% OPEN | 0% OPEN |
| 9 | Fresh Air AHU | OFF | OFF | OFF |

## Sleep Mode :

- Minimum Blower Speed

# Room Temperature Control

A. Control Strategy 1:
   Using warm water.
B. Control Strategy 2:
   Using both warm water and chilled water.

# Room Humidity Control

A. Control strategy 1:

   Comparing air temp. after cooling coil with calculated dew point temp.

B. Control Strategy 2:

   Using Dehumidifier.

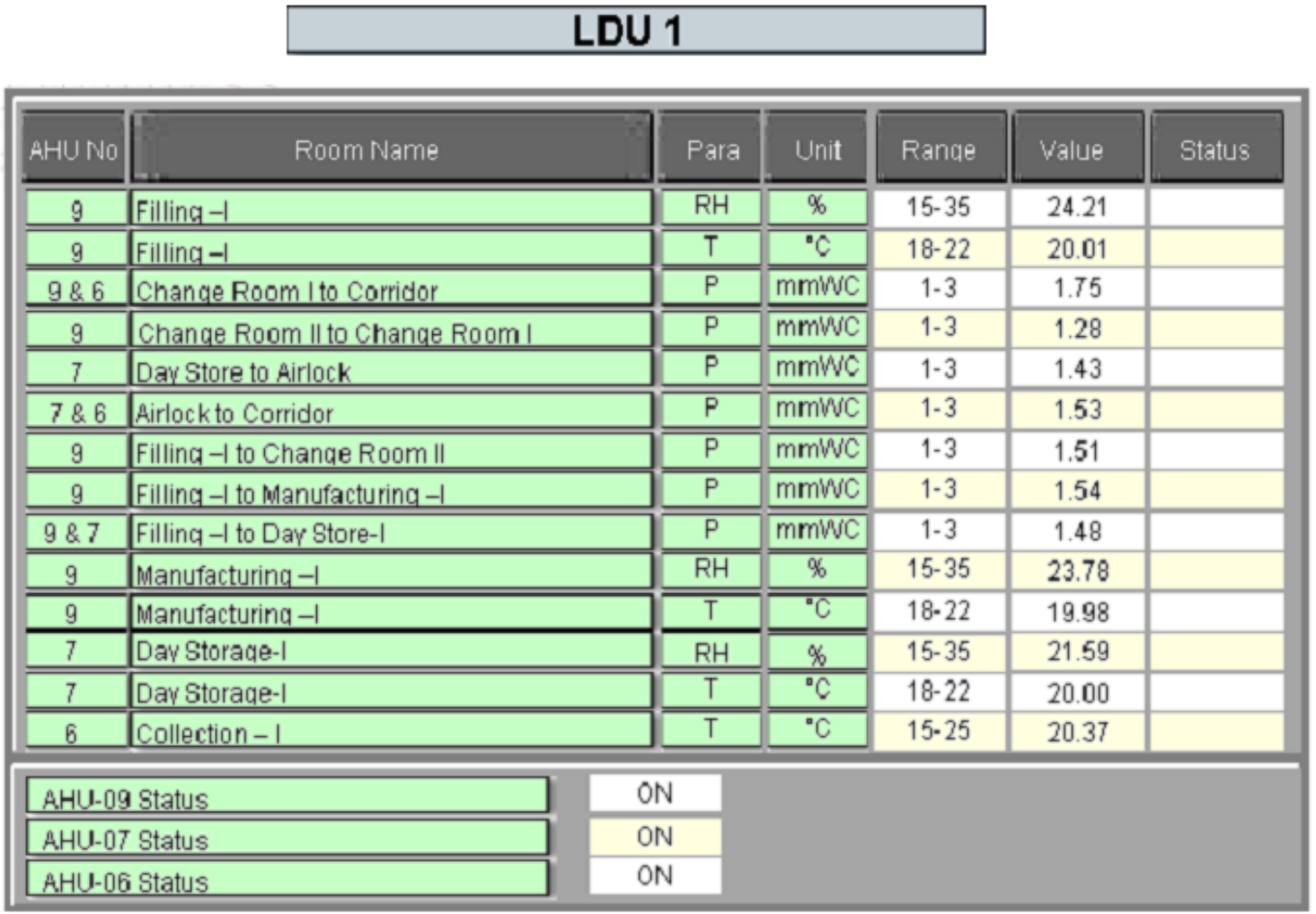
# Room Pressure control

- Modulating VCD in Return Air risers & AHU relief Air Damper.
- Modulating blower speed

# Flasher Control

- Green – ON
- Red – OFF
- Amber – with continuous alarm
- Amber – with cyclic ON/OFF and alarm
- Test
- Accept

# Local Display Unit



**LDU 1**

| AHU No | Room Name | Para | Unit | Range | Value | Status |
|---|---|---|---|---|---|---|
| 9 | Filling –I | RH | % | 15-35 | 24.21 | |
| 9 | Filling –I | T | °C | 18-22 | 20.01 | |
| 9 & 6 | Change Room I to Corridor | P | mmWC | 1-3 | 1.75 | |
| 9 | Change Room II to Change Room I | P | mmWC | 1-3 | 1.28 | |
| 7 | Day Store to Airlock | P | mmWC | 1-3 | 1.43 | |
| 7 & 6 | Airlock to Corridor | P | mmWC | 1-3 | 1.53 | |
| 9 | Filling –I to Change Room II | P | mmWC | 1-3 | 1.51 | |
| 9 | Filling –I to Manufacturing –I | P | mmWC | 1-3 | 1.54 | |
| 9 & 7 | Filling –I to Day Store-I | P | mmWC | 1-3 | 1.48 | |
| 9 | Manufacturing –I | RH | % | 15-35 | 23.78 | |
| 9 | Manufacturing –I | T | °C | 18-22 | 19.98 | |
| 7 | Day Storage-I | RH | % | 15-35 | 21.59 | |
| 7 | Day Storage-I | T | °C | 18-22 | 20.00 | |
| 6 | Collection – I | T | °C | 15-25 | 20.37 | |

| | |
|---|---|
| AHU-09 Status | ON |
| AHU-07 Status | ON |
| AHU-06 Status | ON |

# Power failure recovery

SERVER:

a) UPS provided for at least 10 mins.

b) When power restored and PC turned ON SCADA application restart automatically

Field Controllers :

a) Also provided with UPS backup
b) After power restoration, field controllers restart and execute application data files stored in RAM.

VFD PANEL :

a) Blower motor restart based on last state before loss of power.

# Alarms and Warnings

- CRITICAL
- Priority:

a) Urgent
b) High
    1) PVHI
    2) PVHH
c) Low
    1) PVLO
    2) PVLL

- NON – CRITICAL
- Priority:

a) Journal

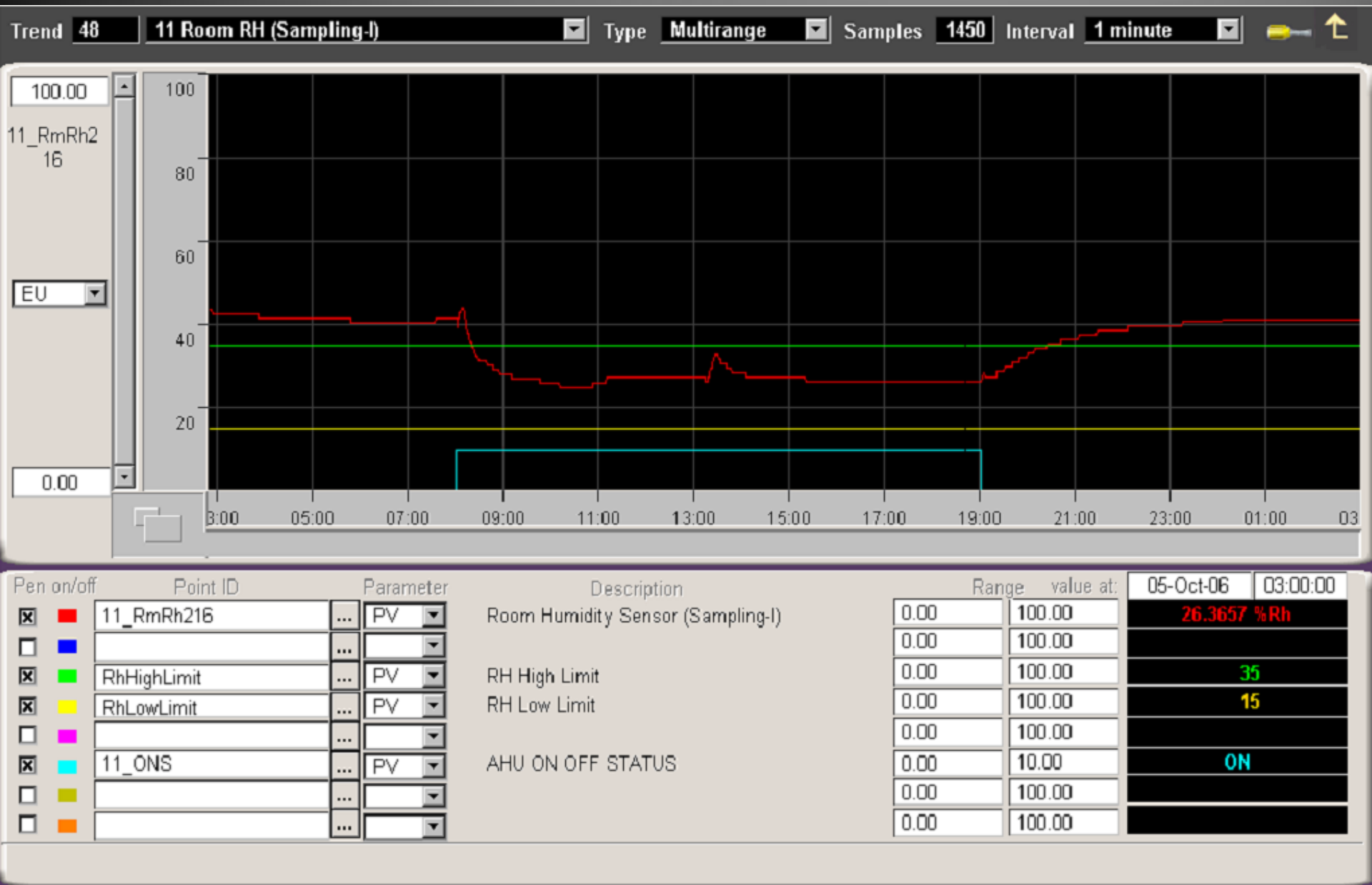
e.g., Flasher ON/OFF,
Report generated

# Security levels

| | |
|---|---|
| LVL1 | View only mode |
| LVL2 | Alarm acknowledgement mode |
| OPER | Operator mode |
| SUPV | Supervisor mode |
| ENGR | Engineer mode |
| MNGR | Manager mode |

You Cannot access any Station functions until you have
Successfully signed on.

# Trend Reporting



# THANK YOU.